Lecture-12
Inviscid Flows III: Applications of the Bernoulli Equation

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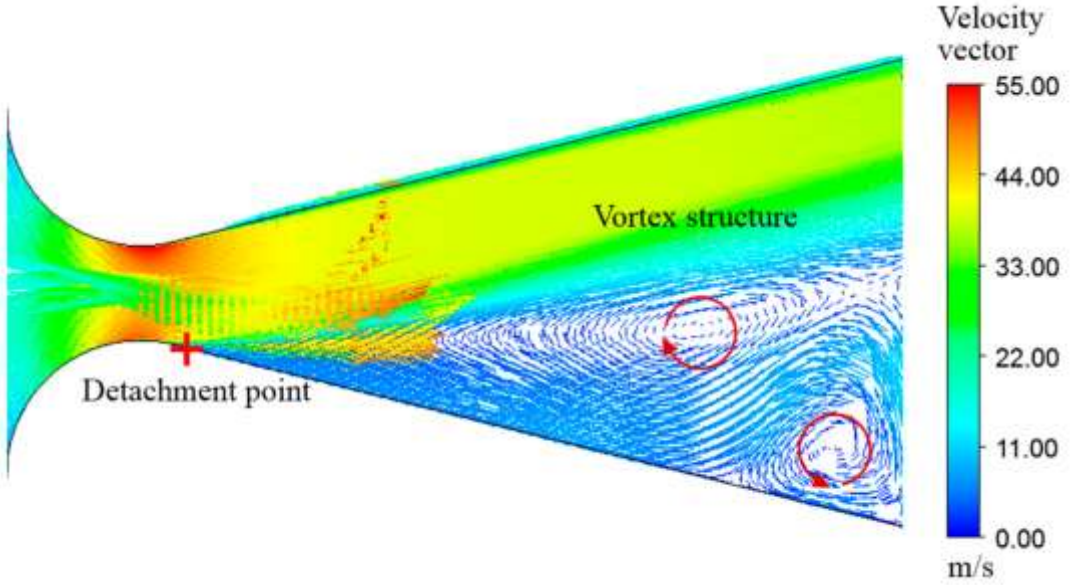
Learning Objectives

By the end of this lecture, you should be able to

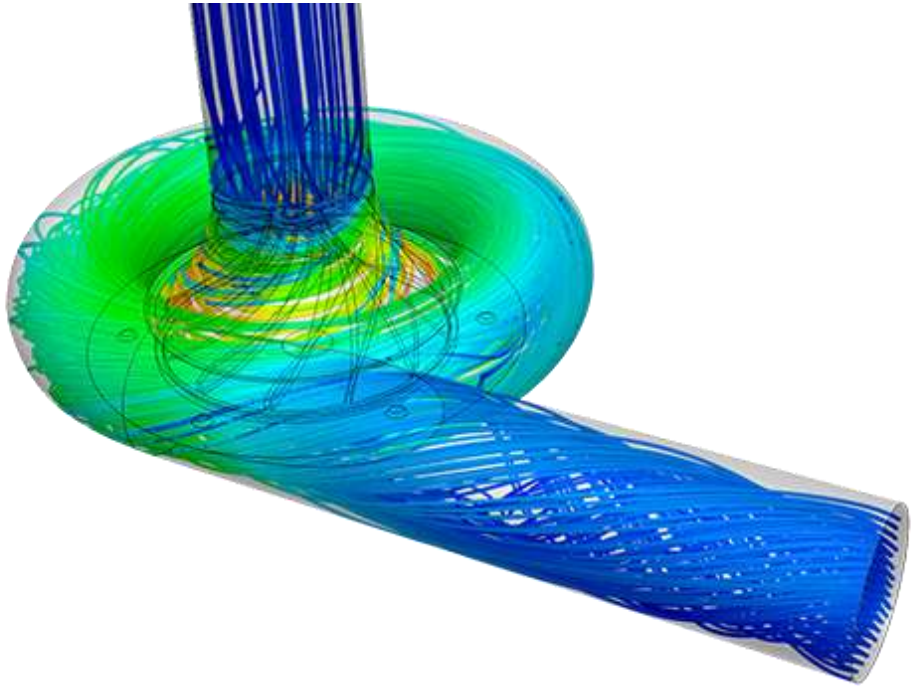
1. Apply the **steady state Bernoulli equation** to practical engineering problems
2. Evaluate the **limitations of the Bernoulli equation** in real-world systems
3. Formulate the **unsteady Bernoulli equation**
4. Differentiate between **Hydraulic Grade Line** and **Energy Grade Line**

Cautions while using Bernoulli Equation

- The central assumptions while deriving the Bernoulli equation, such as **incompressibility**, **inviscid**, and **steady state** conditions, in addition to **applying along a streamline** or **irrotationality**, should be kept in mind.



Subsonic Diffuser



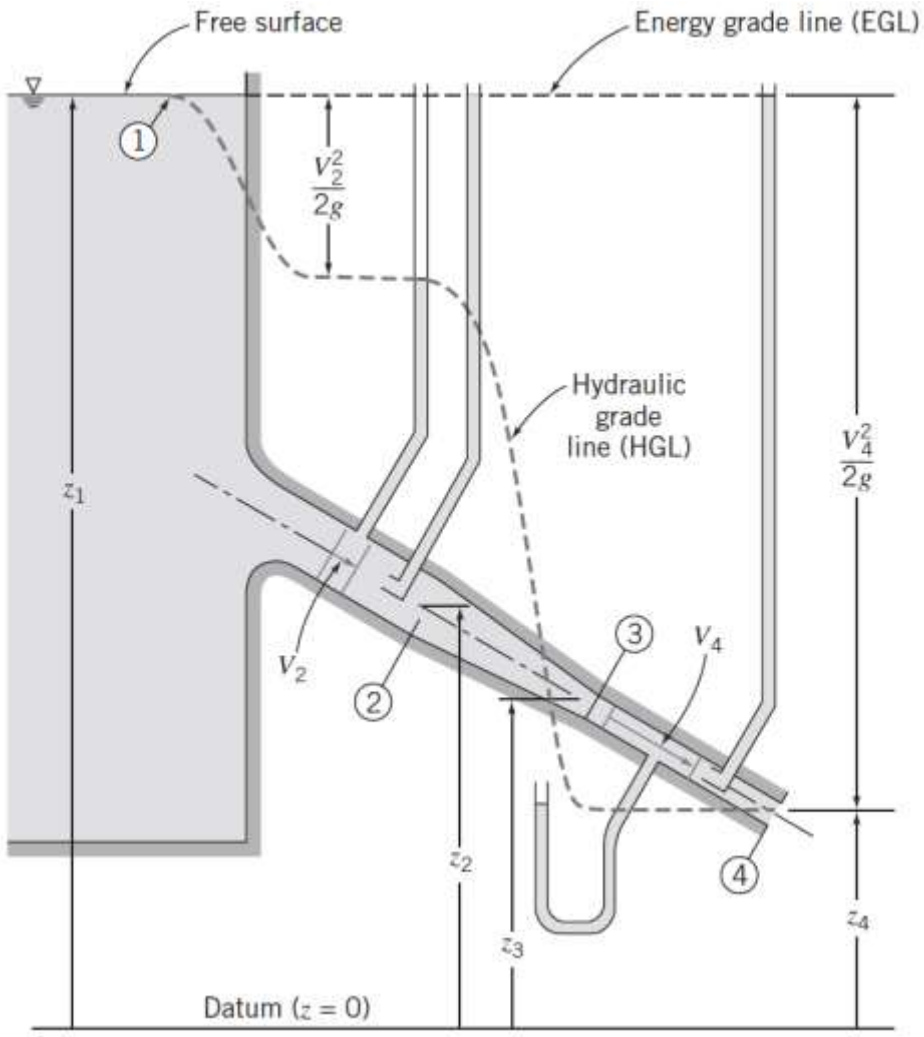
Centrifugal Pump

https://www.mdpi.com/2227-9717/8/2/143
https://www.simscale.com/blog/what-is-centrifugal-pump/

Energy Grade Line and Hydraulic Grade Line

$$EGL = \frac{p}{\rho g} + \frac{V^2}{2g} + z$$

$$HGL = \frac{p}{\rho g} + z$$



Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Unsteady version of the Bernoulli Equation

- What happens if **all assumptions except steady flow** hold in a flow?

$$\frac{p_1}{\rho}+\frac{V_1^2}{2}+gz_1=\frac{p_2}{\rho}+\frac{V_2^2}{2}+gz_2+\int_1^2\frac{\partial V}{\partial t}ds$$

- The additional term represents the **work involved in the rate of increase of momentum of the fluid on the streamline over time**.

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Revisiting the learning objectives

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